

Solutions, HW 1, Part 2

September 11, 2020

```
In [2]: import numpy as np
import pandas as pd
import scipy.stats
from sklearn import linear_model
import statsmodels.api as sm
import matplotlib.pyplot as plt
```

1 Solutions for Homework 1, Part 2

Questions 1 and 2 are worth one point each. The remainder are worth two points each.

2 Q1

```
In [3]: data = pd.read_csv("data.csv")
```

```
In [3]: data.head()
```

```
Out[3]:
```

	PERMNO	date	TICKER	COMNAM	CUSIP	BIDLO	\
0	84415	20000929	CRYS	CRYSTAL SYSTEMS SOLUTIONS	M7037810	7.8750	
1	84415	20001031	CRYS	CRYSTAL SYSTEMS SOLUTIONS	M7037810	5.6250	
2	84415	20001130	CRYS	CRYSTAL SYSTEMS SOLUTIONS	M7037810	5.8750	
3	84415	20001229	CRYS	CRYSTAL SYSTEMS SOLUTIONS	M7037810	7.0625	
4	84415	20010131	CRYS	CRYSTAL SYSTEMS SOLUTIONS	M7037810	6.8750	

	ASKHI	PRC	BID	ASK	SHROUT	CFACPR
0	9.125	8.3125	8.0625	8.4375	7000	0.25
1	8.375	5.6250	5.5000	6.0625	7000	0.25
2	7.750	7.5000	7.2500	7.5000	7000	0.25
3	8.500	7.2500	6.7500	7.1875	7000	0.25
4	8.375	8.1250	8.0625	8.1875	7000	0.25

```
In [4]: data.shape
```

```
Out[4]: (356, 12)
```

There are 356 rows (stock-day combinations) and 12 columns.

3 Q2

```
In [4]: set(data['TICKER'])
```

```
Out[4]: {'BBY', 'BPHX', 'CRYS', 'MDSY'}
```

```
In [5]: set(data['CUSIP'])
```

```
Out[5]: {'08651610', 'M7037810'}
```

There are 4 distinct tickers but only 2 distinct CUSIP IDs, because the ticker symbol for a stock can change over time.

4 Q3

```
In [6]: data[data['CUSIP'] == 'M7037810'].iloc[30:40]
```

```
Out[6]:
```

	PERMNO	date	TICKER	COMNAM	CUSIP	BIDLO	\
30	84415	20030331	CRYS	CRYSTAL SYSTEMS SOLUTIONS	M7037810	-2.160	
31	84415	20030430	CRYS	CRYSTAL SYSTEMS SOLUTIONS	M7037810	2.180	
32	84415	20030530	CRYS	CRYSTAL SYSTEMS SOLUTIONS	M7037810	-2.445	
33	84415	20030630	CRYS	CRYSTAL SYSTEMS SOLUTIONS	M7037810	2.640	
34	84415	20030731	CRYS	CRYSTAL SYSTEMS SOLUTIONS	M7037810	3.050	
35	84415	20030829	BPHX	BLUEPHOENIX SOLUTIONS LTD	M7037810	3.310	
36	84415	20030930	BPHX	BLUEPHOENIX SOLUTIONS LTD	M7037810	3.431	
37	84415	20031031	BPHX	BLUEPHOENIX SOLUTIONS LTD	M7037810	3.750	
38	84415	20031128	BPHX	BLUEPHOENIX SOLUTIONS LTD	M7037810	4.250	
39	84415	20031231	BPHX	BLUEPHOENIX SOLUTIONS LTD	M7037810	5.200	

	ASKHI	PRC	BID	ASK	SHROUT	CFACPR
30	2.41	2.300	2.18	2.30	13449	0.25
31	2.55	-2.445	2.41	2.48	13449	0.25
32	2.70	2.510	2.50	2.70	13449	0.25
33	3.25	-3.075	3.05	3.10	13449	0.25
34	3.70	3.510	3.51	3.70	13449	0.25
35	3.69	3.540	3.37	3.54	13449	0.25
36	4.00	3.750	3.75	3.84	13449	0.25
37	4.55	4.250	4.25	4.54	13449	0.25
38	5.50	5.030	4.86	5.03	13449	0.25
39	7.04	5.900	5.96	5.99	13484	0.25

We see that the ticker symbol changed the same month that the company changed its name. We confirm this with a google search: <https://www.businesswire.com/news/home/20030810005010/en/Crystal-Systems-Solutions-New-Corporate-BluePhoenix-Solutions>

5 Q4

```
In [7]: set(data['CFACPR'])
```

```
Out[7]: {0.25, 1.0, 1.5, 2.25}
```

There are 4 values for CFACPR in the dataset.

```
In [8]: data[data['CFACPR'].isna()]
```

```
Out[8]: Empty DataFrame
```

```
Columns: [PERMNO, date, TICKER, COMNAM, CUSIP, BIDLO, ASKHI, PRC, BID, ASK, SHROUT, CFACPR]
```

```
Index: []
```

There are no missing values for CFACPR.

```
In [9]: data['PRCadj'] = data['PRC'] / data['CFACPR']
```

```
In [10]: data[data['CFACPR'] == 1.5].iloc[0]
```

```
Out[10]: PERMNO      85914
         date      20020531
         TICKER      BBY
         COMNAM  BEST BUY COMPANY INC
         CUSIP      08651610
         BIDLO      45.05
         ASKHI      74.8
         PRC        46.2
         BID        46.07
         ASK        46.28
         SHROUT     318694
         CFACPR      1.5
         PRCadj     30.8
         Name: 198, dtype: object
```

In May 2002, the price of Best Buy was \$\$\$46.20 but there was an adjustment factor of 1.5, so the adjusted price was \$\$\$30.80.

6 Q5

```
In [11]: data[data['PRCadj'].isna()]
```

```
Out[11]: Empty DataFrame
```

```
Columns: [PERMNO, date, TICKER, COMNAM, CUSIP, BIDLO, ASKHI, PRC, BID, ASK, SHROUT, CFA
```

```
Index: []
```

There are no “not available” adjusted prices.

7 Q6

```
In [12]: len(set(data['date']))
```

```
Out[12]: 178
```

There are 178 unique dates in the dataset.

8 Q7

```
In [13]: data[data['PRC'] < 0]
```

```
Out[13]:
```

	PERMNO	date	TICKER	COMNAM	CUSIP	BIDLO	\
19	84415	20020430	CRYS	CRYSTAL SYSTEMS SOLUTIONS	M7037810	3.4100	
29	84415	20030228	CRYS	CRYSTAL SYSTEMS SOLUTIONS	M7037810	2.1000	
31	84415	20030430	CRYS	CRYSTAL SYSTEMS SOLUTIONS	M7037810	2.1800	
33	84415	20030630	CRYS	CRYSTAL SYSTEMS SOLUTIONS	M7037810	2.6400	
58	84415	20050729	BPHX	BLUEPHOENIX SOLUTIONS LTD	M7037810	3.8500	
71	84415	20060831	BPHX	BLUEPHOENIX SOLUTIONS LTD	M7037810	4.6000	
164	84415	20140530	BPHX	BLUEPHOENIX SOLUTIONS LTD	M7037810	4.1000	
168	84415	20140930	BPHX	BLUEPHOENIX SOLUTIONS LTD	M7037810	3.5501	
172	84415	20150130	MDSY	MODSYS INTERNATIONAL LTD	M7037810	3.0500	
175	84415	20150430	MDSY	MODSYS INTERNATIONAL LTD	M7037810	2.2601	
177	84415	20150630	MDSY	MODSYS INTERNATIONAL LTD	M7037810	1.7800	

	ASKHI	PRC	BID	ASK	SHROUT	CFACPR	PRCadj
19	4.00	-3.585	3.40	3.77	10018	0.25	-14.340
29	2.30	-2.165	2.13	2.20	13449	0.25	-8.660
31	2.55	-2.445	2.41	2.48	13449	0.25	-9.780
33	3.25	-3.075	3.05	3.10	13449	0.25	-12.300
58	4.25	-4.050	4.00	4.10	13556	0.25	-16.200
71	5.70	-5.500	5.45	5.55	13560	0.25	-22.000
164	4.30	-4.100	4.05	4.15	11468	1.00	-4.100
168	3.81	-3.595	3.50	3.69	11621	1.00	-3.595
172	3.46	-3.075	2.95	3.20	17844	1.00	-3.075
175	3.10	-2.620	2.29	2.95	17897	1.00	-2.620
177	2.13	-1.830	1.80	1.86	17926	1.00	-1.830

Negative prices in the CRSP dataset indicate that the closing price was missing for that date, so instead the average of the bid and ask prices were used. We note that by inspecting the volume data that these stocks were quite lightly traded during these months.

```
In [15]: data.loc[data['PRC'] < 0, 'PRC'] = -data[data['PRC'] < 0]['PRC']
         data.loc[data['PRCadj'] < 0, 'PRCadj'] = -data[data['PRCadj'] < 0]['PRCadj']
```

```
In [17]: data[data['PRC'] < 0]
```

```
Out[17]: Empty DataFrame
```

```
Columns: [PERMNO, date, TICKER, COMNAM, CUSIP, BIDLO, ASKHI, PRC, BID, ASK, SHROUT, CFA]
Index: []
```

9 Q8

```
In [17]: data[data['BID'] > data['ASK']]
```

```
Out[17]:
```

	PERMNO	date	TICKER	COMNAM	CUSIP	BIDLO	ASKHI	\
252	85914	20061130	BBY	BEST BUY COMPANY INC	08651610	51.39	55.84	

	PRC	BID	ASK	SHROUT	CFACPR	PRCadj
252	54.97	54.92	54.91	480250	1.0	54.97

In Nov 2006, the bid price of Best Buy was higher than the ask. This could be a data error, or the closing bid entered at the last minute and the market closed before any trade was able to be resolved. Instances of a so-called **crossed market** can happen in times of rapid trading. See <https://www.investopedia.com/terms/c/crossedmarket.asp>.

10 Q9

```
In [18]: bby = data.loc[data['CUSIP'] == '08651610'].copy()
        modsys = data.loc[data['CUSIP'] == 'M7037810'].copy()
```

```
In [19]: bby['momentum'] = bby['PRCadj'].shift(1) / bby['PRCadj'].shift(4) - 1
        modsys['momentum'] = modsys['PRCadj'].shift(1) / modsys['PRCadj'].shift(4) - 1
```

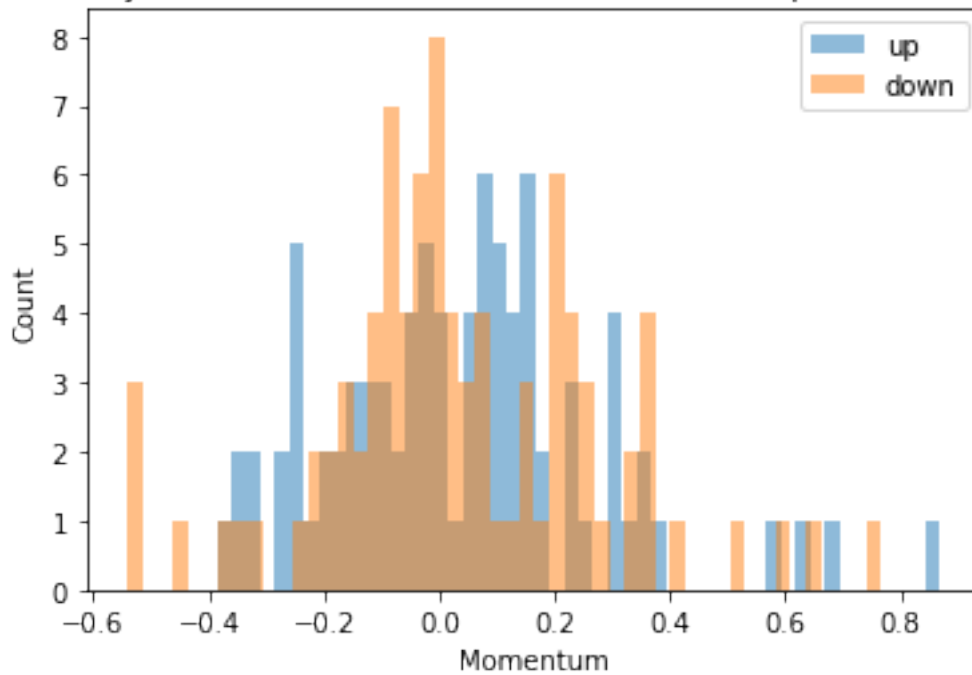
11 Q10

```
In [19]: bby['PRCinc'] = bby['PRCadj'].shift(-1) > bby['PRCadj']
        bby['PRCdec'] = bby['PRCadj'].shift(-1) < bby['PRCadj']
        modsys['PRCinc'] = modsys['PRCadj'].shift(-1) > modsys['PRCadj']
        modsys['PRCdec'] = modsys['PRCadj'].shift(-1) < modsys['PRCadj']

In [20]: plt.hist(bby[bby['PRCinc'] == 1]['momentum'], bins=50, alpha=0.5, label="up")
        plt.hist(bby[bby['PRCdec'] == 1]['momentum'], bins=50, alpha=0.5, label="down")
        plt.legend(loc='upper right')
        plt.title("Best Buy: Distribution of Momentum Scores for Up or Down Moves")
        plt.xlabel("Momentum")
        plt.ylabel("Count")
        plt.show()
```

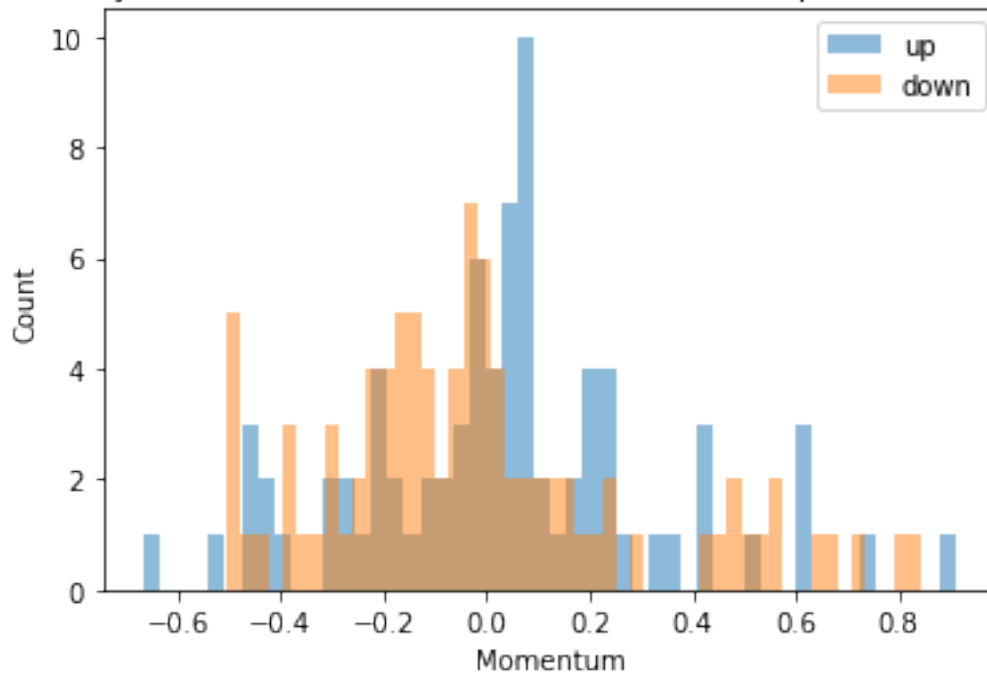
```
/Users/chadschafer/anaconda3/lib/python3.7/site-packages/numpy/lib/histograms.py:824: RuntimeWarning:
  keep = (tmp_a >= first_edge)
/Users/chadschafer/anaconda3/lib/python3.7/site-packages/numpy/lib/histograms.py:825: RuntimeWarning:
  keep &= (tmp_a <= last_edge)
```

Best Buy: Distribution of Momentum Scores for Up or Down Moves



```
In [21]: plt.hist(modsys[modsys['PRCinc'] == 1]['momentum'], bins=50, alpha=0.5, label="up")
plt.hist(modsys[modsys['PRCdec'] == 1]['momentum'], bins=50, alpha=0.5, label="down")
plt.legend(loc='upper right')
plt.title("ModSys: Distribution of Momentum Scores for Up or Down Moves")
plt.xlabel("Momentum")
plt.ylabel("Count")
plt.show()
```

ModSys: Distribution of Momentum Scores for Up or Down Moves



The distribution of momentum scores is slightly more positive (shifted to the right) when the next month return is positive, as compared to the distribution of momentum scores when the next month return is negative.